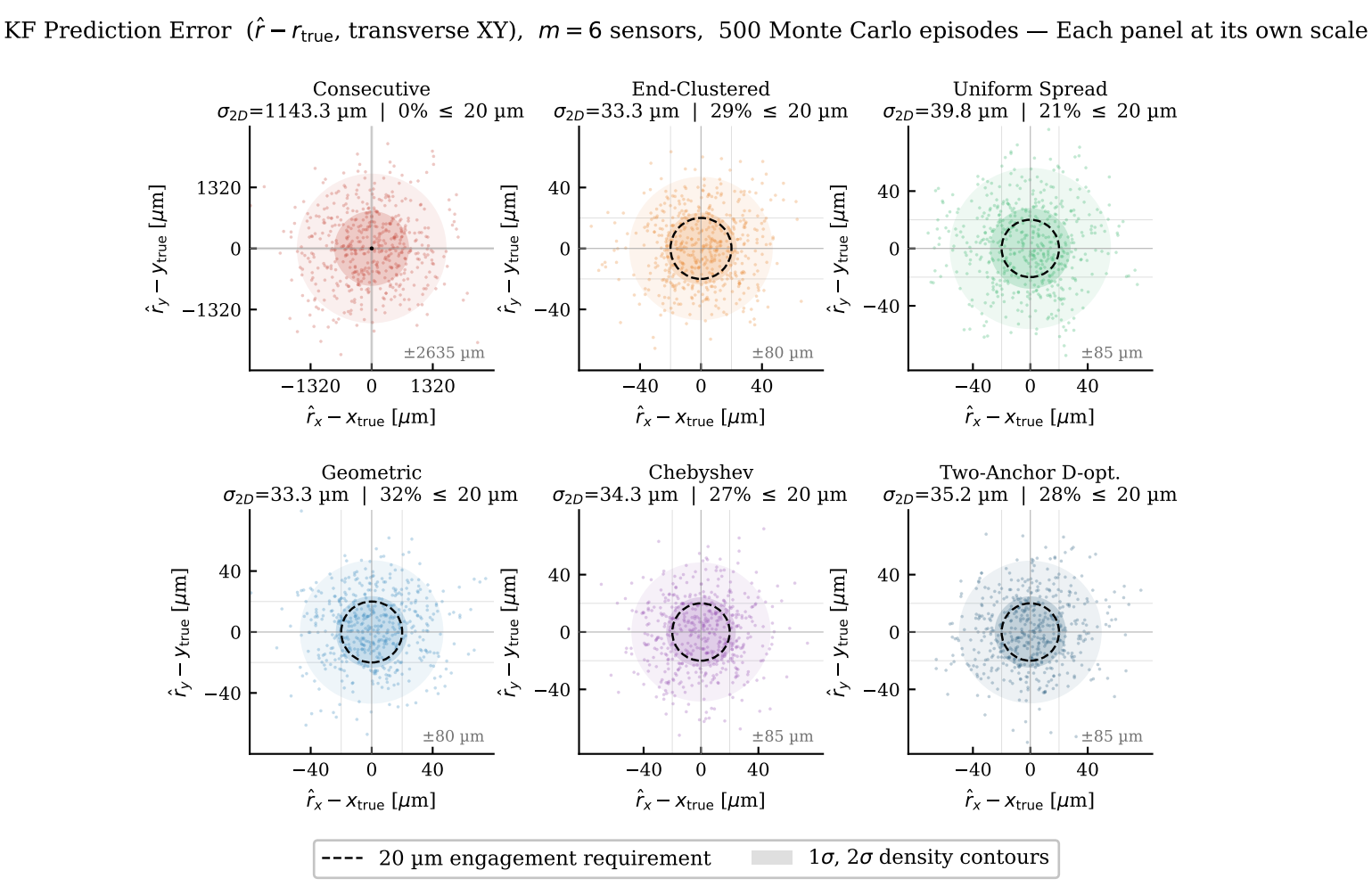

KF Prediction Error ($\hat{r} - r_{\text{true}}$, transverse XY), $m = 6$ sensors, 500 Monte Carlo episodes — Each panel at its own scale
Consecutive
σ_{2D}=1143.3 µm | 0% ≤ 20 µm
End-Clustered
σ_{2D}=33.3 µm | 29% ≤ 20 µm
Uniform Spread
σ_{2D}=39.8 µm | 21% ≤ 20 µm
Geometric
σ_{2D}=33.3 µm | 32% ≤ 20 µm
Chebyshev
σ_{2D}=34.3 µm | 27% ≤ 20 µm
Two-Anchor D-opt.
σ_{2D}=35.2 µm | 28% ≤ 20 µm
$\hat{r}_y - y_{\text{true}}$ [µm]
$\hat{r}_x - x_{\text{true}}$ [µm]
±2635 µm
±80 µm
±85 µm
20 µm engagement requirement
1σ, 2σ density contours